MSG No. 191.238 $P6'/m'm'm$ [Type III, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `6'/m'm'm`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: `1b`, site symmetry: `6'/m'm'm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: `2c`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 5: Wyckoff site: `2e`, site symmetry: `6'm'm`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$[0, 0, -z]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 6: Wyckoff site: `3f`, site symmetry: `m'm'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,5,7,11,14,16,20,22]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,6,8,12,15,17,21,23]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,4,9,10,13,18,19,24]

Table 7: Wyckoff site: `3g`, site symmetry: `m'm'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,5,7,11,14,16,20,22]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,6,8,12,15,17,21,23]

3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,9,10,13,18,19,24]`

Table 8: Wyckoff site: `4h`, site symmetry: `3m'`.

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,22,23,24]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[4,5,6,19,20,21]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9,16,17,18]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[10,11,12,13,14,15]`

Table 9: Wyckoff site: `6i`, site symmetry: `2'm'm`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,11,14,22]`
2	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,12,15,23]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[3,10,13,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,9,18,19]`
5	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,7,16,20]`
6	$\left[0,\ \frac{1}{2},\ -z\right]$	`[6,8,17,21]`

Table 10: Wyckoff site: `6j`, site symmetry: `m'2'm`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,11,16,20]`
2	$[0,\ x,\ 0]$	`[2,12,17,21]`
3	$[-x,\ -x,\ 0]$	`[3,10,18,19]`
4	$[x,\ x,\ 0]$	`[4,9,13,24]`
5	$[-x,\ 0,\ 0]$	`[5,7,14,22]`
6	$[0,\ -x,\ 0]$	`[6,8,15,23]`

Table 11: Wyckoff site: `6k`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,11,16,20]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,12,17,21]`
3	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[3,10,18,19]`
4	$\left[x,\ x,\ \frac{1}{2}\right]$	`[4,9,13,24]`
5	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[5,7,14,22]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,8,15,23]`

Table 12: Wyckoff site: `6l`, site symmetry: `m'm'2`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,5,20,22]`
2	$[-2x,\ -x,\ 0]$	`[2,6,21,23]`
3	$[x,\ -x,\ 0]$	`[3,4,19,24]`
4	$[-x,\ -2x,\ 0]$	`[7,11,14,16]`
5	$[2x,\ x,\ 0]$	`[8,12,15,17]`
6	$[-x,\ x,\ 0]$	`[9,10,13,18]`

Table 13: Wyckoff site: `6m`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,5,20,22]`
2	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[2,6,21,23]`
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,4,19,24]`
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[7,11,14,16]`
5	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[8,12,15,17]`
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,10,13,18]`

Table 14: Wyckoff site: `12n`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,11]`
2	$[0,\ x,\ z]$	`[2,12]`
3	$[-x,\ -x,\ z]$	`[3,10]`
4	$[x,\ x,\ -z]$	`[4,9]`
5	$[-x,\ 0,\ -z]$	`[5,7]`
6	$[0,\ -x,\ -z]$	`[6,8]`
7	$[x,\ x,\ z]$	`[13,24]`
8	$[-x,\ 0,\ z]$	`[14,22]`
9	$[0,\ -x,\ z]$	`[15,23]`
10	$[x,\ 0,\ -z]$	`[16,20]`
11	$[0,\ x,\ -z]$	`[17,21]`
12	$[-x,\ -x,\ -z]$	`[18,19]`

Table 15: Wyckoff site: `12o`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,22]`
2	$[-2x,\ -x,\ z]$	`[2,23]`
3	$[x,\ -x,\ z]$	`[3,24]`
4	$[x,\ -x,\ -z]$	`[4,19]`
5	$[x,\ 2x,\ -z]$	`[5,20]`
6	$[-2x,\ -x,\ -z]$	`[6,21]`
7	$[-x,\ -2x,\ -z]$	`[7,16]`

continued ...

Table 15

No.	position	mapping
8	$[2x,\ x,\ -z]$	[8,17]
9	$[-x,\ x,\ -z]$	[9,18]
10	$[-x,\ x,\ z]$	[10,13]
11	$[-x,\ -2x,\ z]$	[11,14]
12	$[2x,\ x,\ z]$	[12,15]

Table 16: Wyckoff site: 12p, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,20]
2	$[-y,\ x-y,\ 0]$	[2,21]
3	$[-x+y,\ -x,\ 0]$	[3,19]
4	$[x,\ x-y,\ 0]$	[4,24]
5	$[-x+y,\ y,\ 0]$	[5,22]
6	$[-y,\ -x,\ 0]$	[6,23]
7	$[-x,\ -y,\ 0]$	[7,14]
8	$[y,\ -x+y,\ 0]$	[8,15]
9	$[x-y,\ x,\ 0]$	[9,13]
10	$[-x,\ -x+y,\ 0]$	[10,18]
11	$[x-y,\ -y,\ 0]$	[11,16]
12	$[y,\ x,\ 0]$	[12,17]

Table 17: Wyckoff site: 12q, site symmetry: m'..

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,20]
2	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[2,21]
3	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[3,19]
4	$\left[x,\ x-y,\ \frac{1}{2}\right]$	[4,24]
5	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	[5,22]
6	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[6,23]
7	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[7,14]
8	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	[8,15]
9	$\left[x-y,\ x,\ \frac{1}{2}\right]$	[9,13]
10	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	[10,18]
11	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	[11,16]
12	$\left[y,\ x,\ \frac{1}{2}\right]$	[12,17]

Table 18: Wyckoff site: 24r, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]

continued ...

Table 18

No.	position	mapping
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x,\ x-y,\ -z]$	[4]
5	$[-x+y,\ y,\ -z]$	[5]
6	$[-y,\ -x,\ -z]$	[6]
7	$[-x,\ -y,\ -z]$	[7]
8	$[y,\ -x+y,\ -z]$	[8]
9	$[x-y,\ x,\ -z]$	[9]
10	$[-x,\ -x+y,\ z]$	[10]
11	$[x-y,\ -y,\ z]$	[11]
12	$[y,\ x,\ z]$	[12]
13	$[x-y,\ x,\ z]$	[13]
14	$[-x,\ -y,\ z]$	[14]
15	$[y,\ -x+y,\ z]$	[15]
16	$[x-y,\ -y,\ -z]$	[16]
17	$[y,\ x,\ -z]$	[17]
18	$[-x,\ -x+y,\ -z]$	[18]
19	$[-x+y,\ -x,\ -z]$	[19]
20	$[x,\ y,\ -z]$	[20]
21	$[-y,\ x-y,\ -z]$	[21]
22	$[-x+y,\ y,\ z]$	[22]
23	$[-y,\ -x,\ z]$	[23]
24	$[x,\ x-y,\ z]$	[24]